

SPAIII

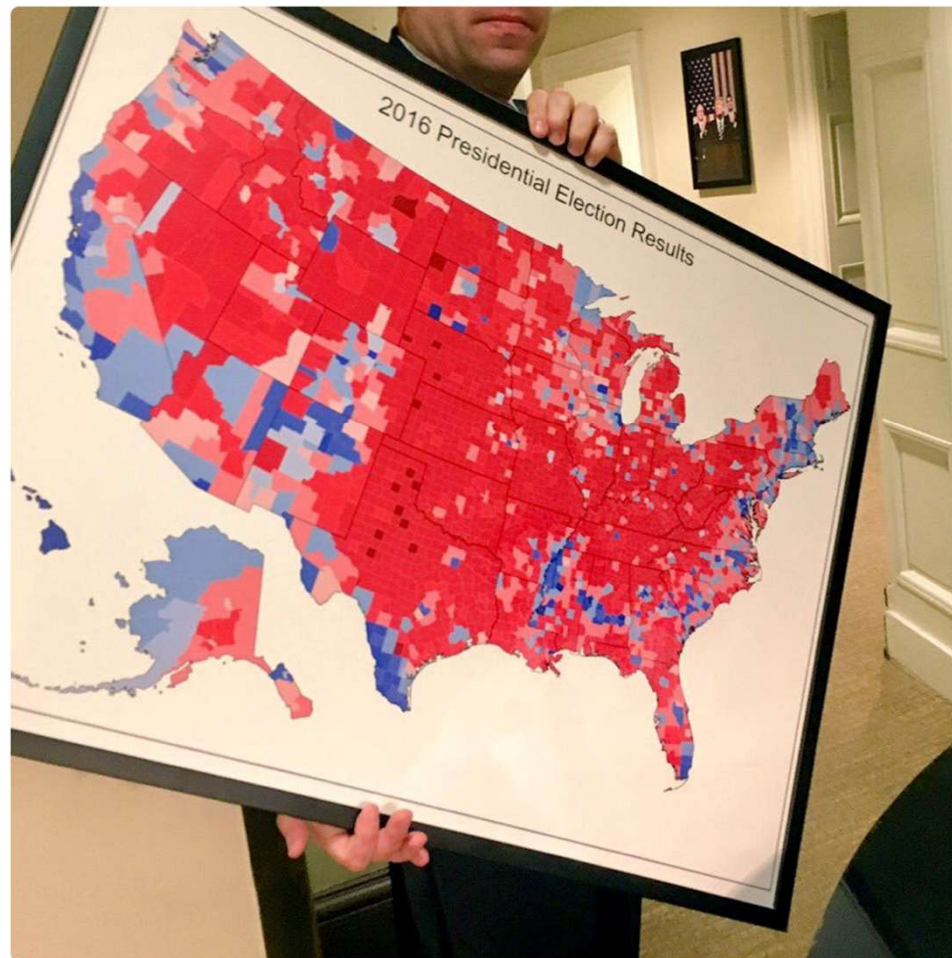


Trey Yingst ✓

@TreyYingst



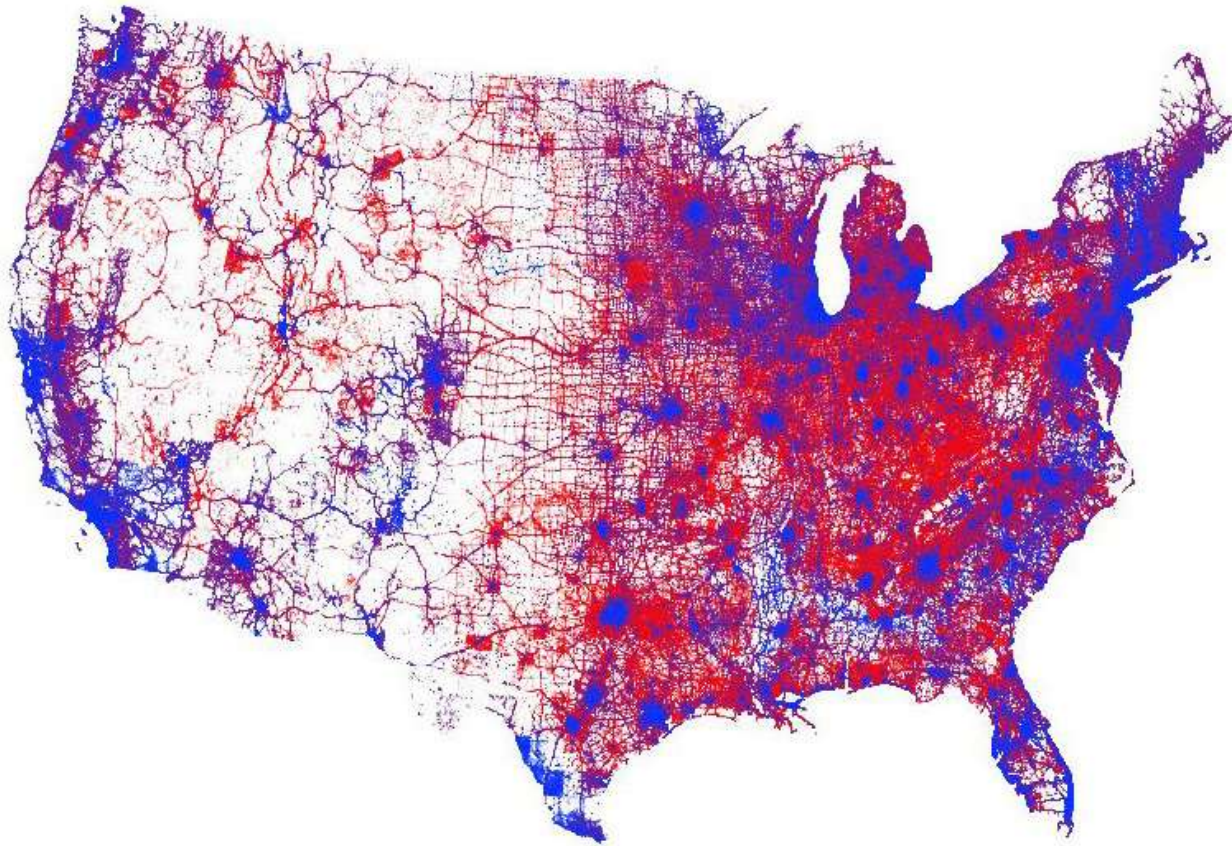
Spotted: A map to be hung somewhere in the West Wing



♥ 8,209 9:03 AM - May 11, 2017



A dasymetric dot map of the same data

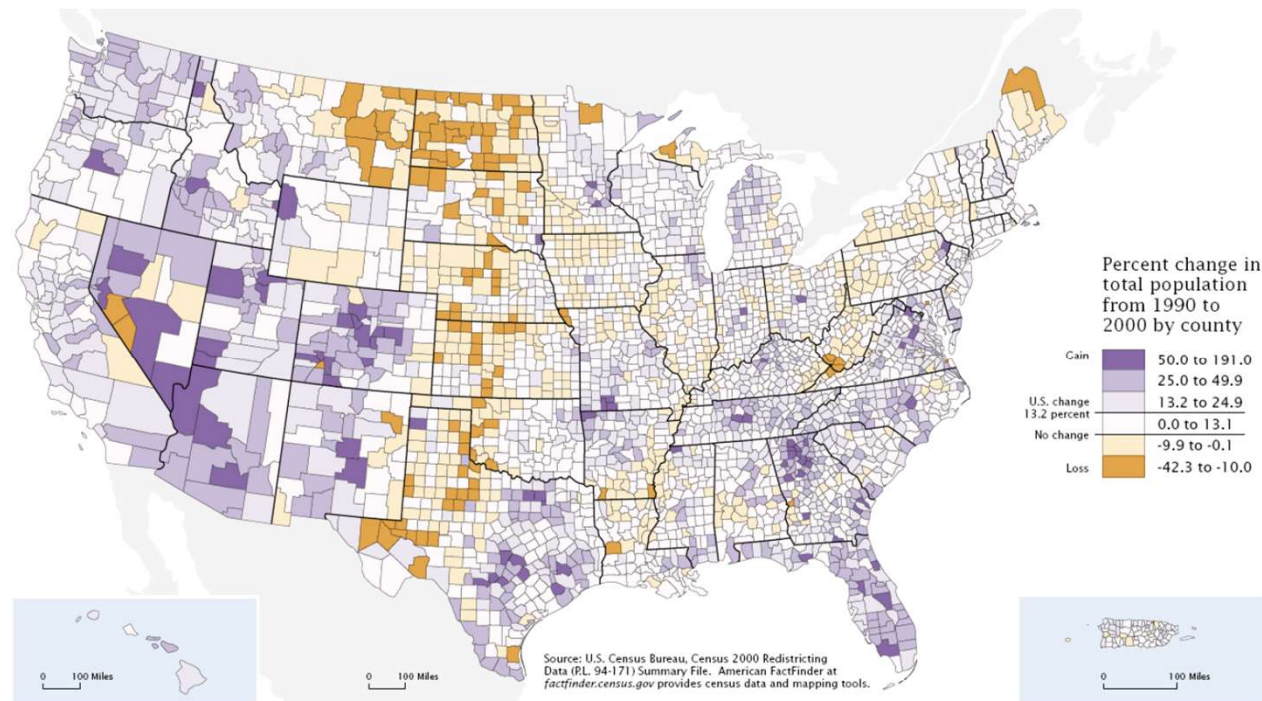


65,844,954 blue dots
62,979,636 red dots

<https://www.esri.com/en-us/maps-we-love/gallery/election-2016-dot-density>

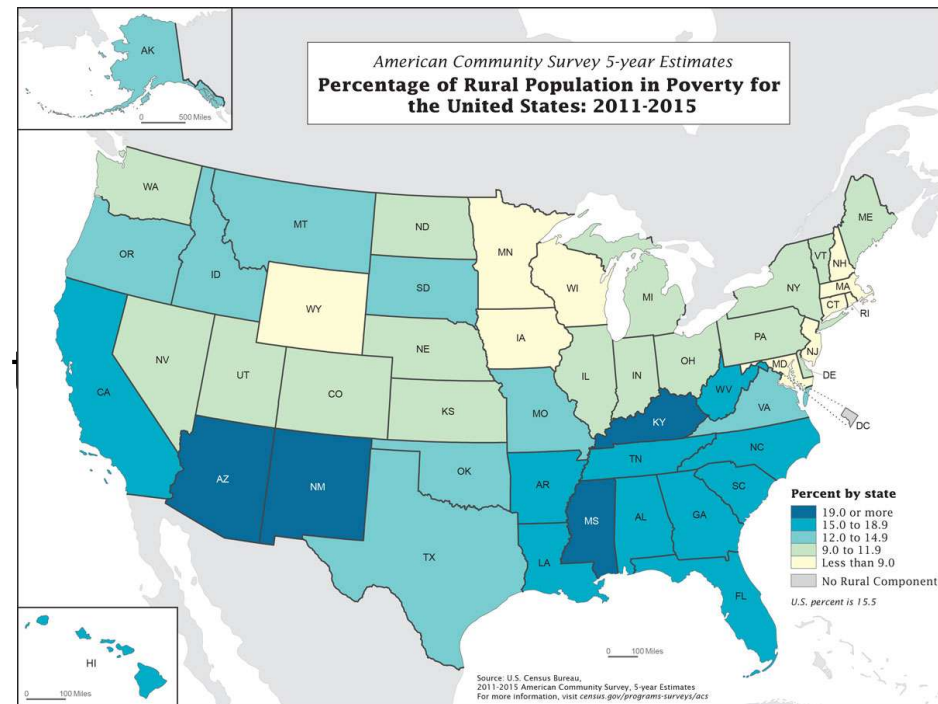
Choropleth maps

- A **choropleth map** displays derived data with colors. The derived data usually are classified prior to mapping. **The appearance of a choropleth map can be greatly affected by data classification**



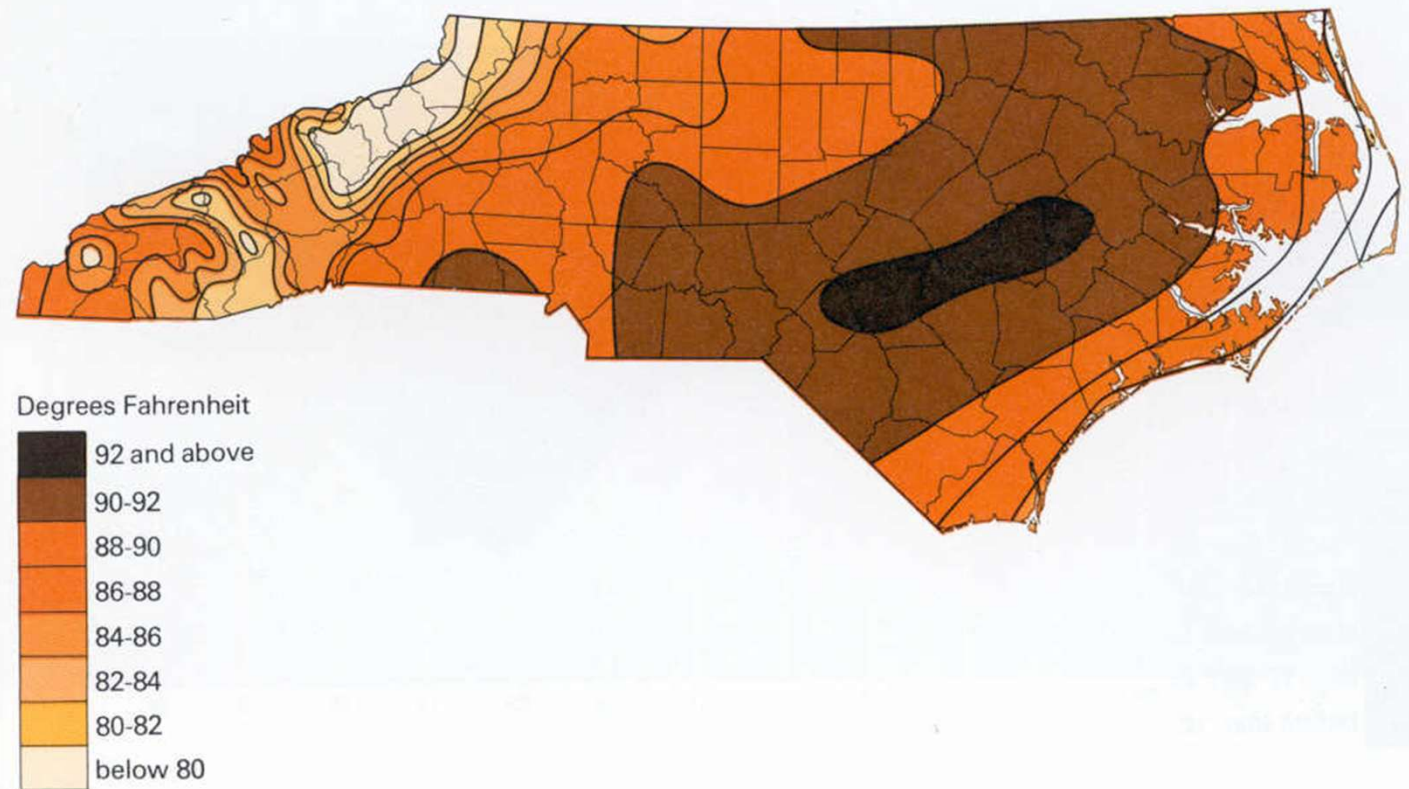
Choropleth maps: design considerations

- Full range of data should be used and classes should not overlap
- Color shades should be easily distinguishable
- Limit number of breaks:
 - 5 classes is best (20% gradient)
 - 3-7 is okay
- Next week we'll discuss methods for



Isarithmic (Isopleth/Isoline) maps

Figure 5.10. Mean Maximum July Temperature in N.C.



Dot maps

Dot value and size

- Dot size too small – washed out map
- Dot size too large – course map
- Single dot value too large – sparse map
- Single dot value too low – super density



Dot maps

- Can be used with all data types, but best for raw counts
 - But changes to color/symbol can show multiple data sets
- Area of aggregation can change appearance of map
- Must consider dot size, value, and arrangement to avoid perception issues

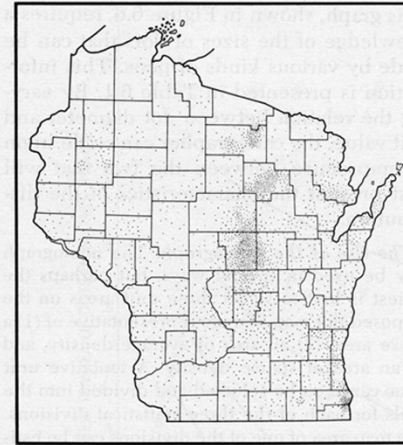


Figure 6.1 A dot map in which the dots are too small so that an unrevealing map is produced. Each dot represents 40 acres in potatoes in 1947.

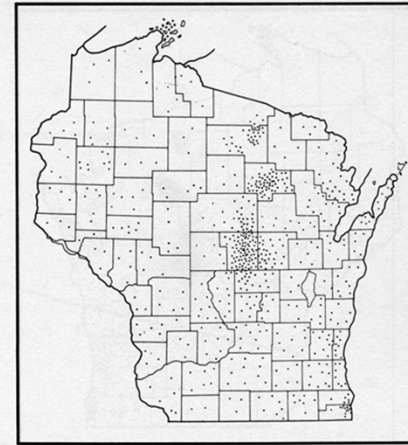


Figure 6.3 A dot map in which the unit value of the dot is too large so that too few dots result; a barren map revealing little pattern is produced. Each dot in this example represents 150 acres.

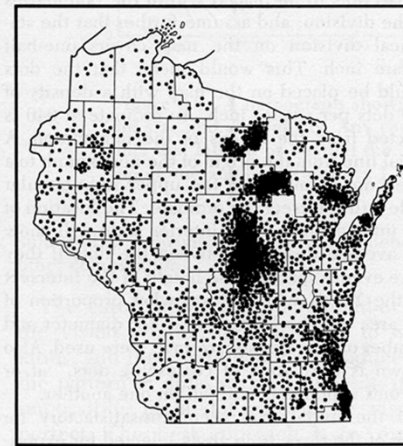


Figure 6.2 A dot map in which the dots are too large so that an excessively "heavy" map is produced. An erroneous impression of excessive potato production is given. The same data and number of dots are used as in Fig. 6.1.

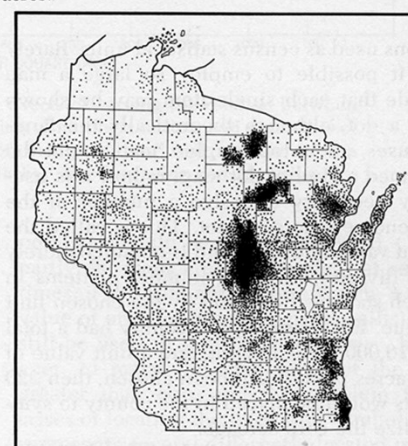


Figure 6.4 A dot map in which the unit value of the dot is too small so that too many dots result; an excessively detailed map is produced. The dots are the same size as those in Fig. 6.3. Each dot in this example represents 15 acres.

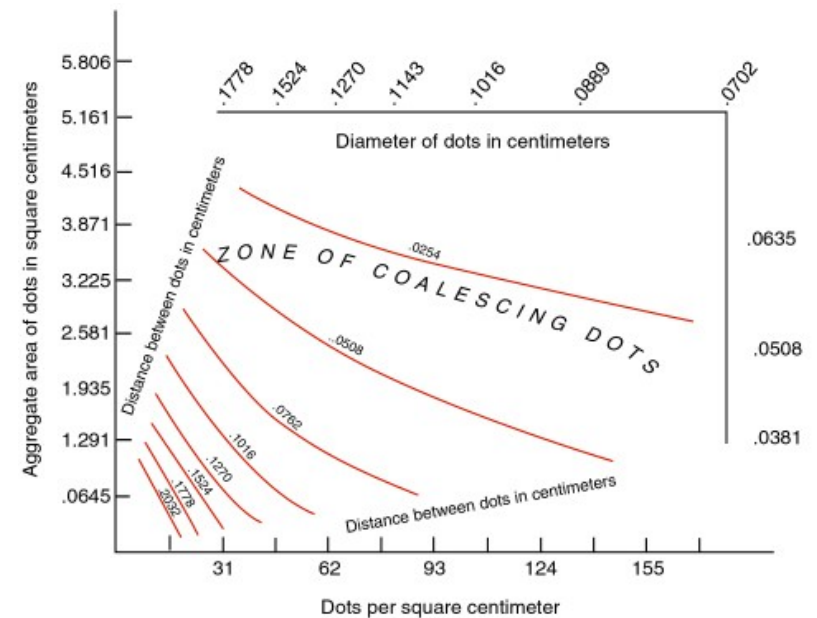
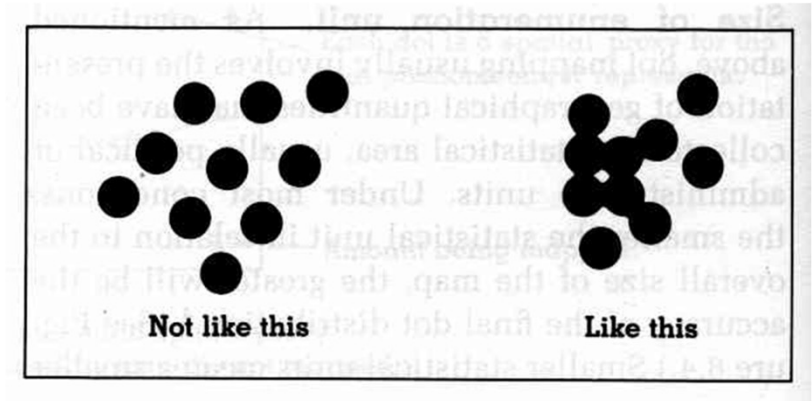
Dot maps

Dot value and size

- Choose size and value so dots just begin to overlap in most dense area

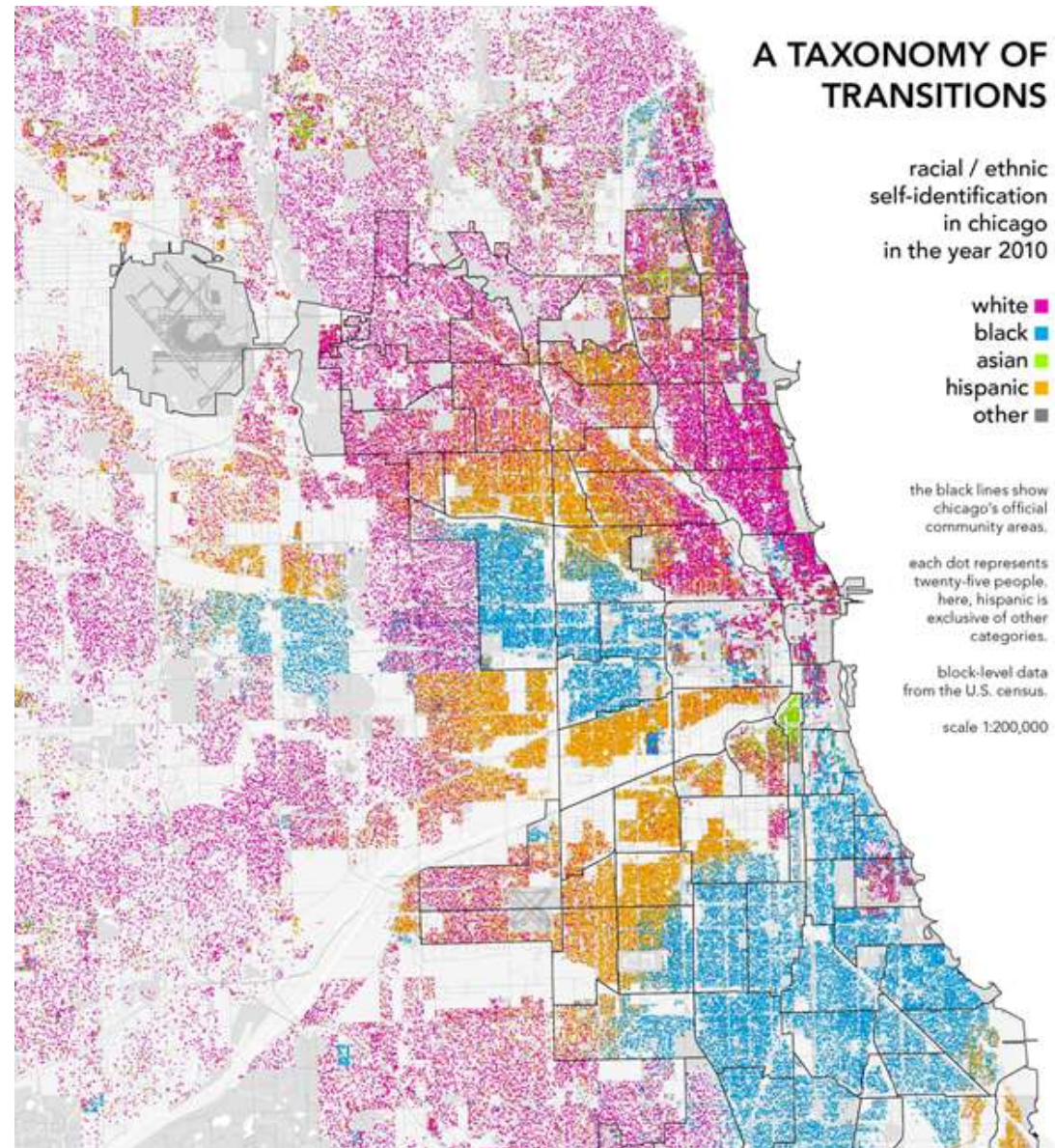
Or

- Use nomograph (MacKay, 1949) to help decide



Dot maps

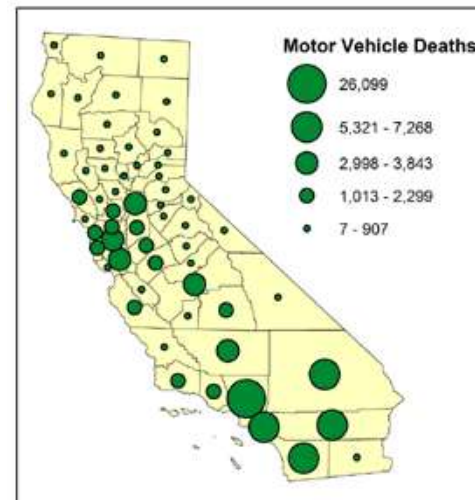
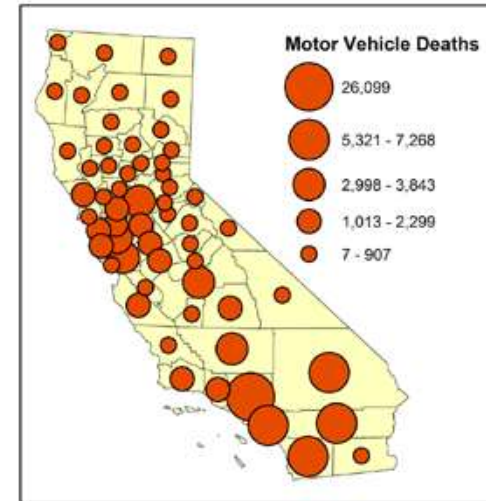
- Multivariate example



<http://www.radicalcartography.net/index.html?chicagodots>

Graduated symbols

- Range-graded symbols
- Easier to discern differences
- More subjective

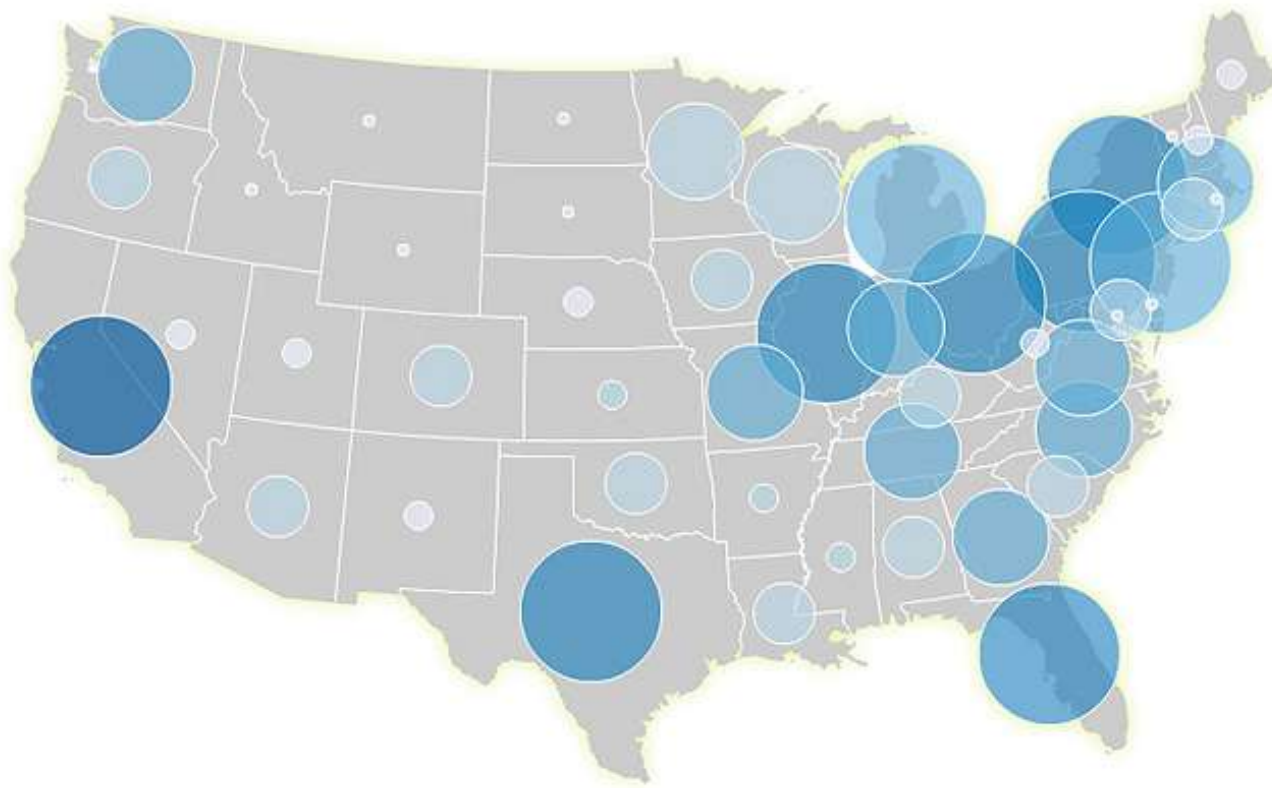


Proportional symbols

- Legend design is one option
- Scales with value

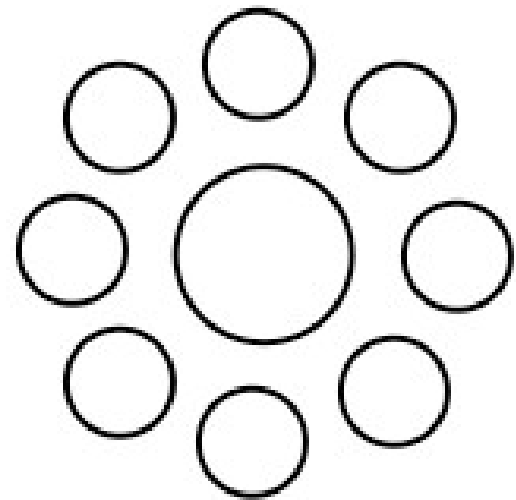
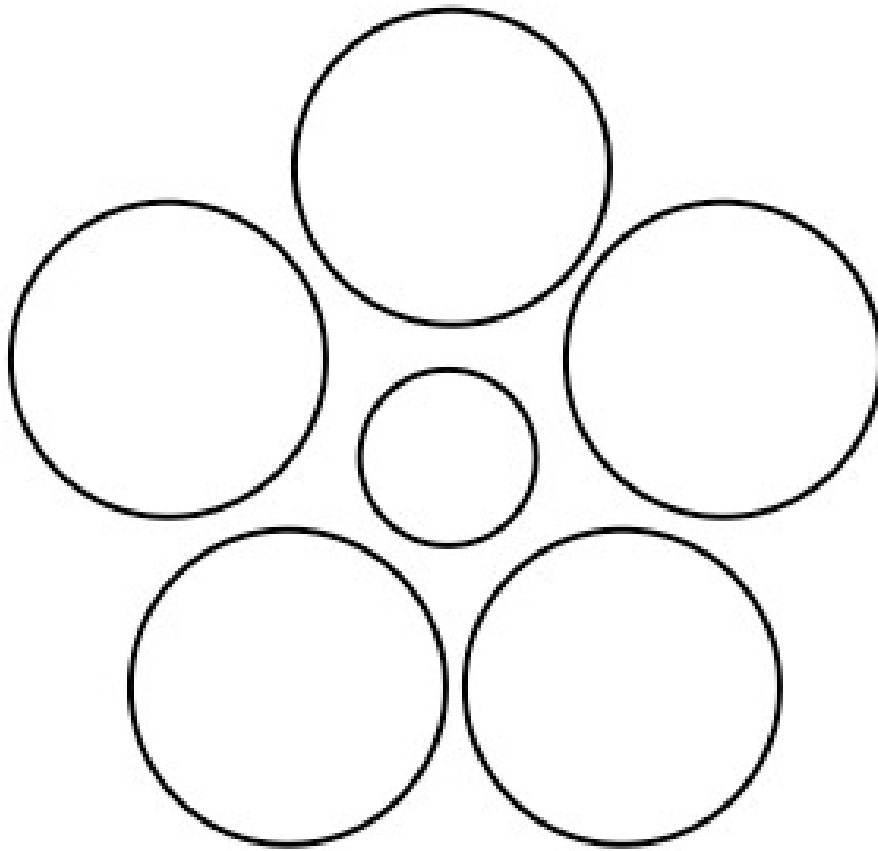


A multivariate proportional symbol map



<https://www.axismaps.com/guide/multivariate/bivariate-proportional-symbols/>

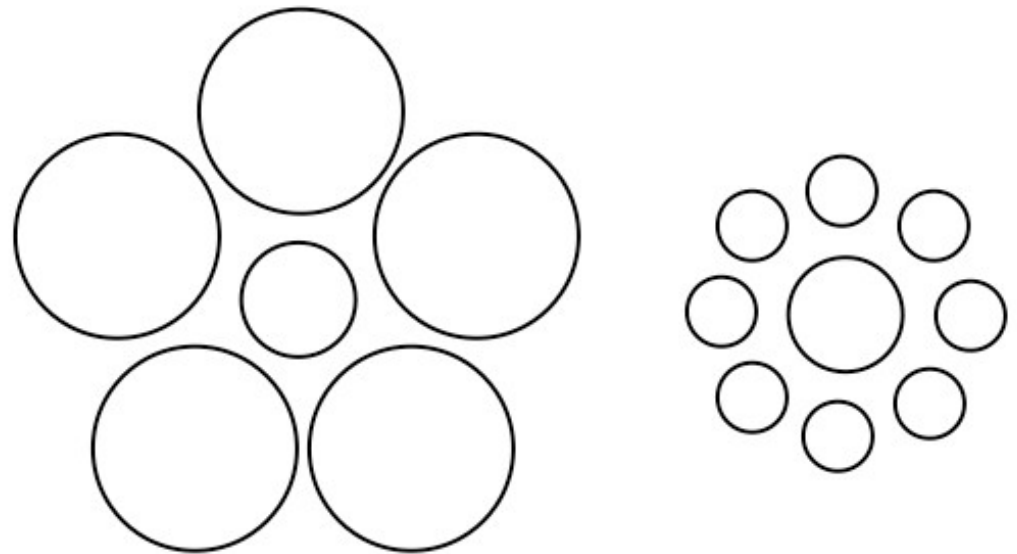
Symbols and perception



Issues of perception

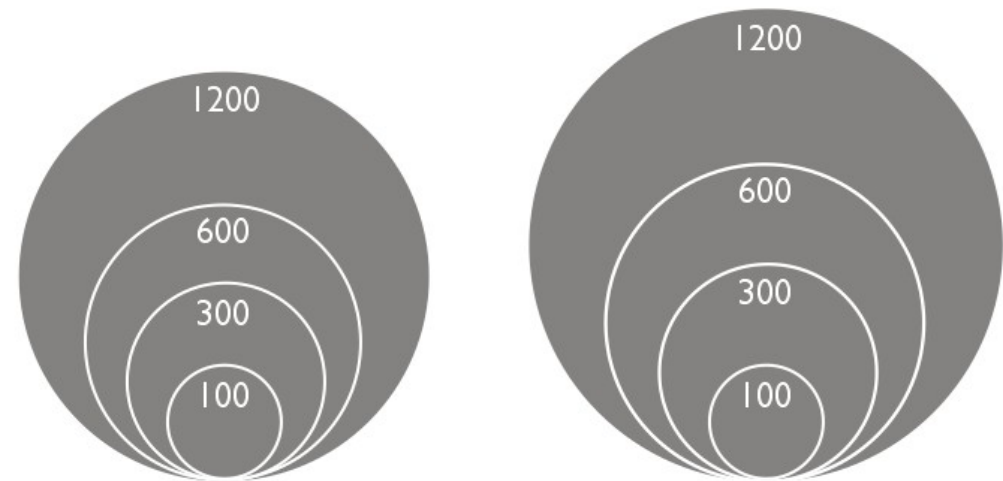
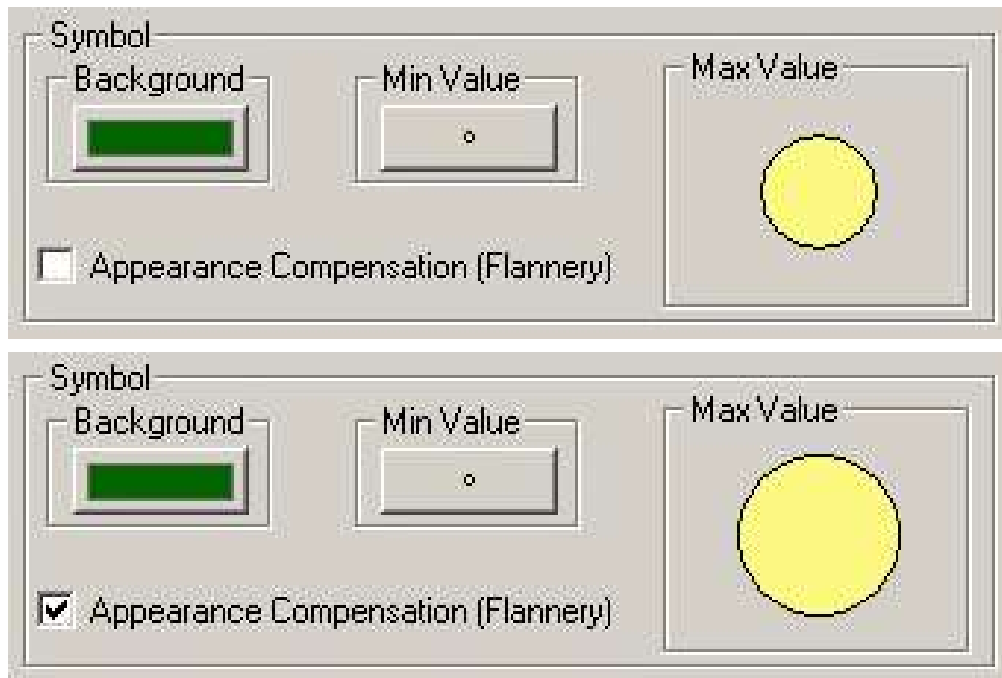
- Humans underestimate the area of proportional symbols
- Especially relative to *other* nearby symbols

A classic example:



Issues of perception: compensating

- So we can compensate



Absolute Scaling

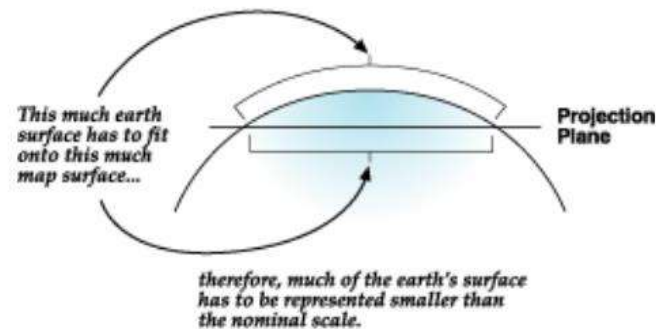
Apparent Scaling
(Flannery's Compensation)

Map projection properties

- Every map projection distorts at least three, and sometimes all four, of the following properties:

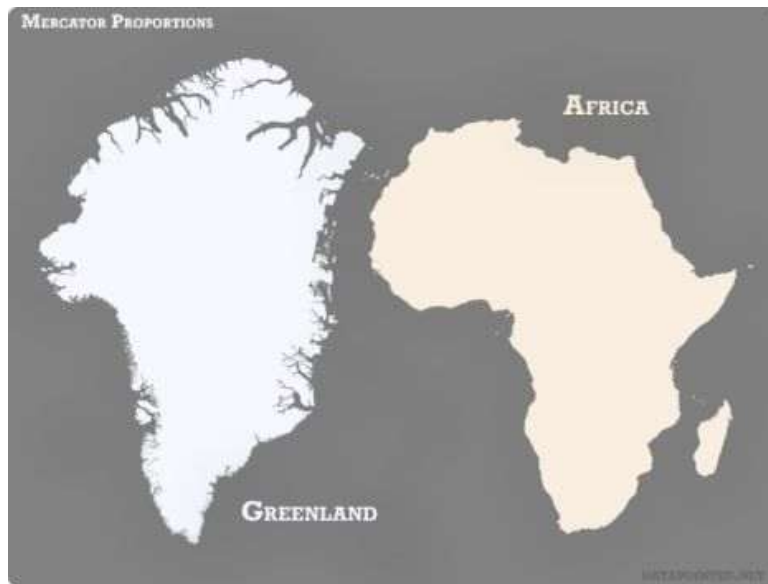
- **Shape**
- **Area**
- **Distance**
- **Direction**

Affect each other



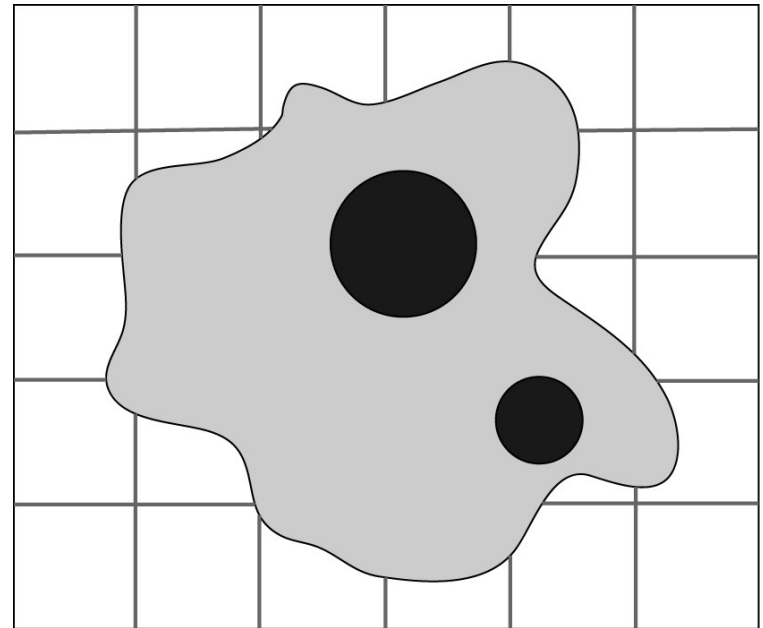
Conformal v. Equivalent

- Comparing Greenland and Africa in two projections, Mercator (conformal, “shape”) and Equivalent projection (true area)



Visual hierarchy

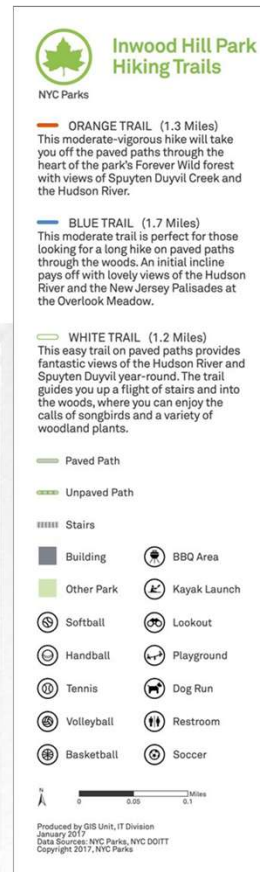
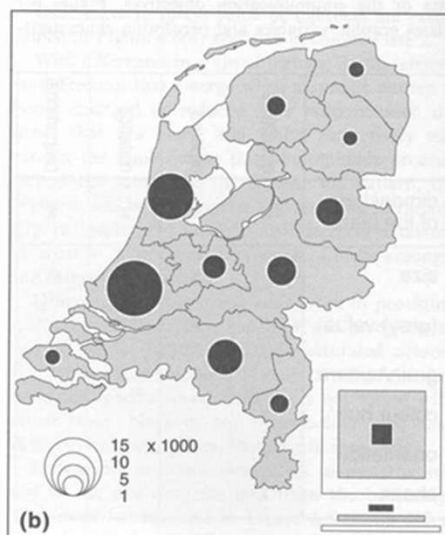
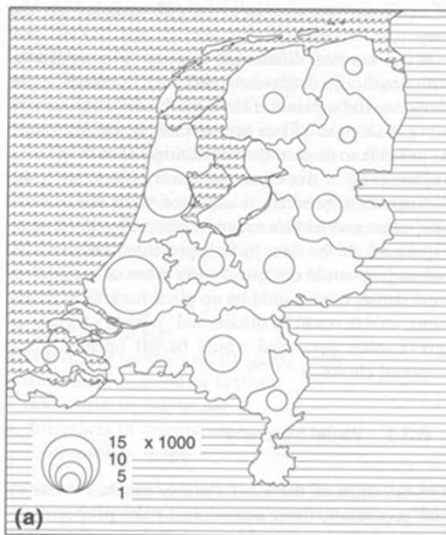
- Visual hierarchy is the process of developing a visual plan to introduce **depth** (or a 3D effect) to maps



A visual hierarchy example. The two black circles are on top (closest to the map reader), followed by the gray polygon and the grid.

Map design: visual hierarchy

- Establish context
- Establish figure-ground relationships



Improving figure-ground

- Can only show the enclosed area (if a polygon)



Improving figure-ground

- Can shade the area



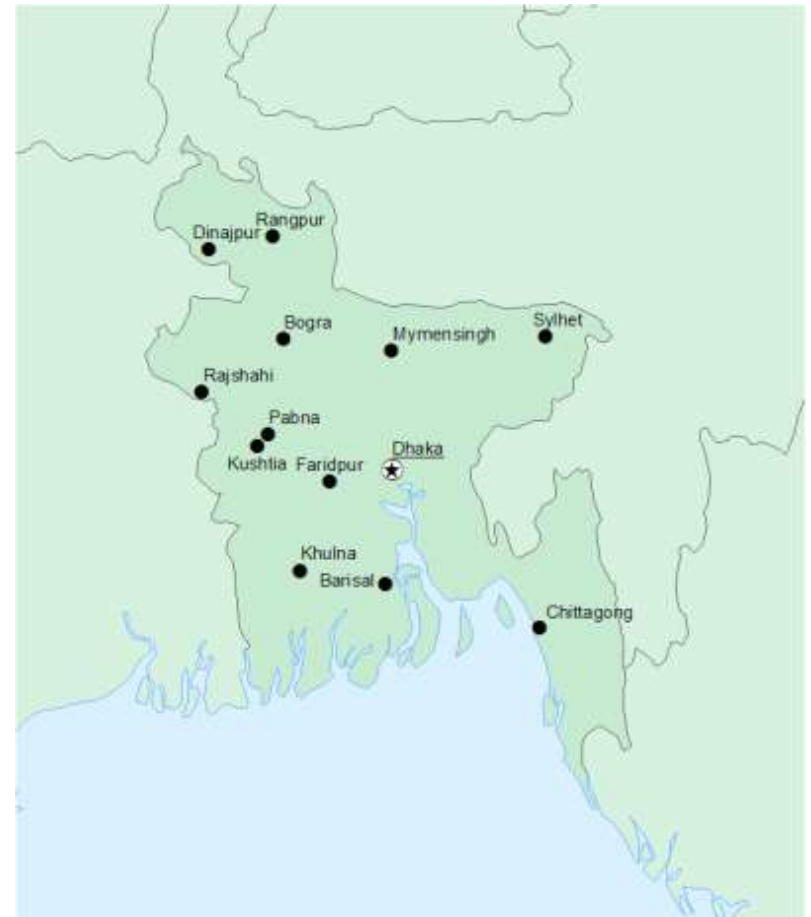
Improving figure-ground

- Can differentiate with other features (by color)



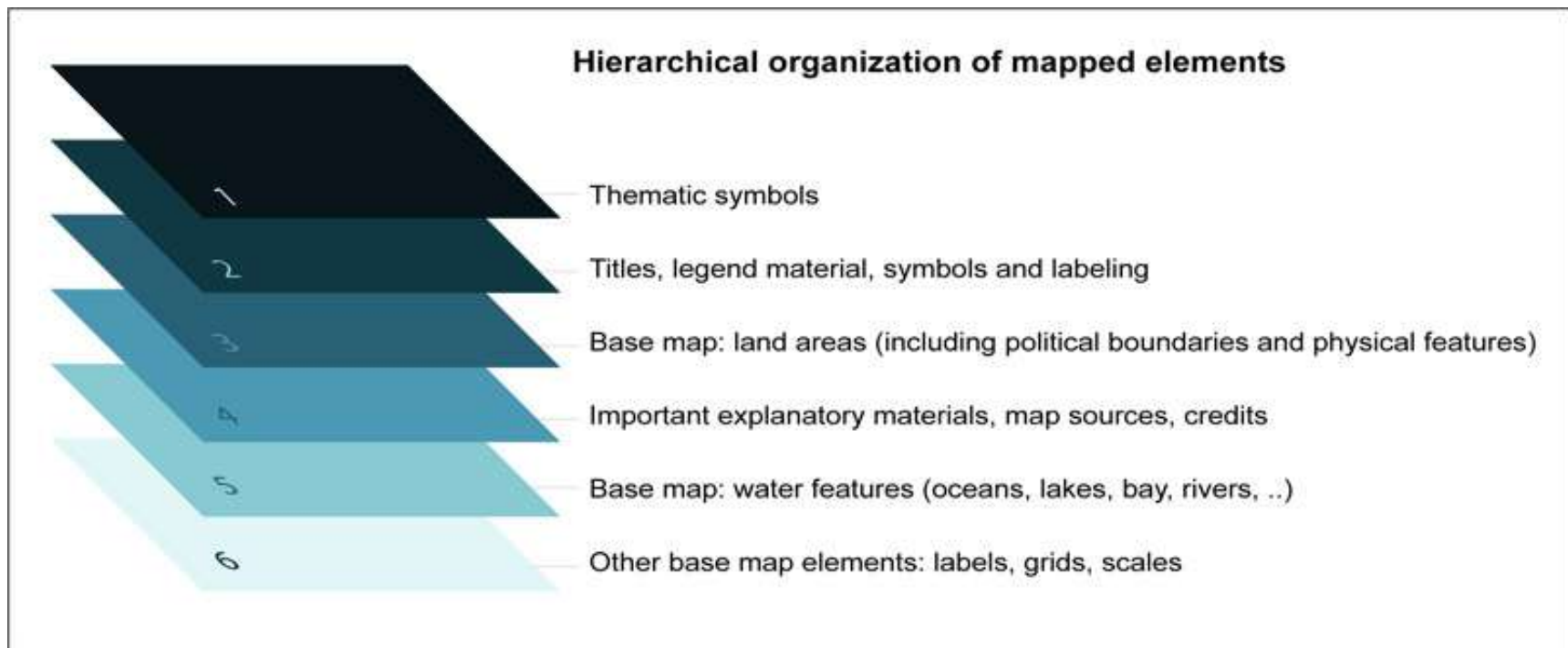
Improving figure-ground

- Can differentiate with other features



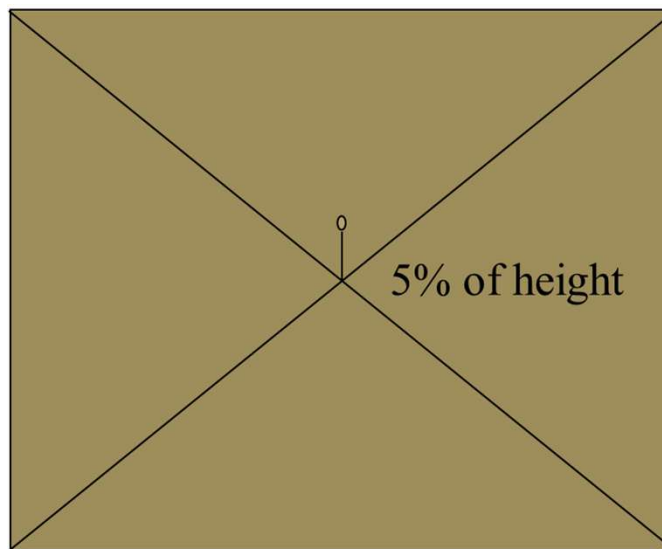
Map design: visual hierarchy

- Determine what elements are the most important!

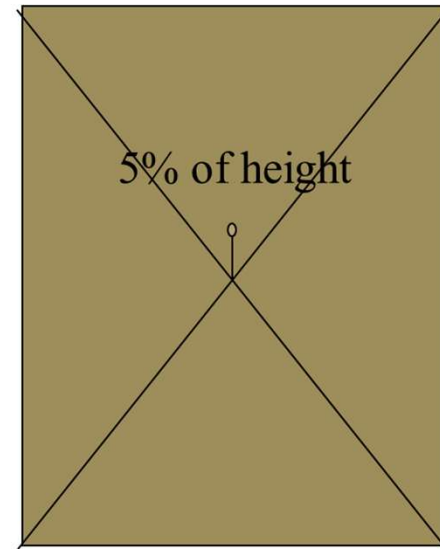


Map design: visual balance

- Center objects around optical center instead of geometric center



Landscape

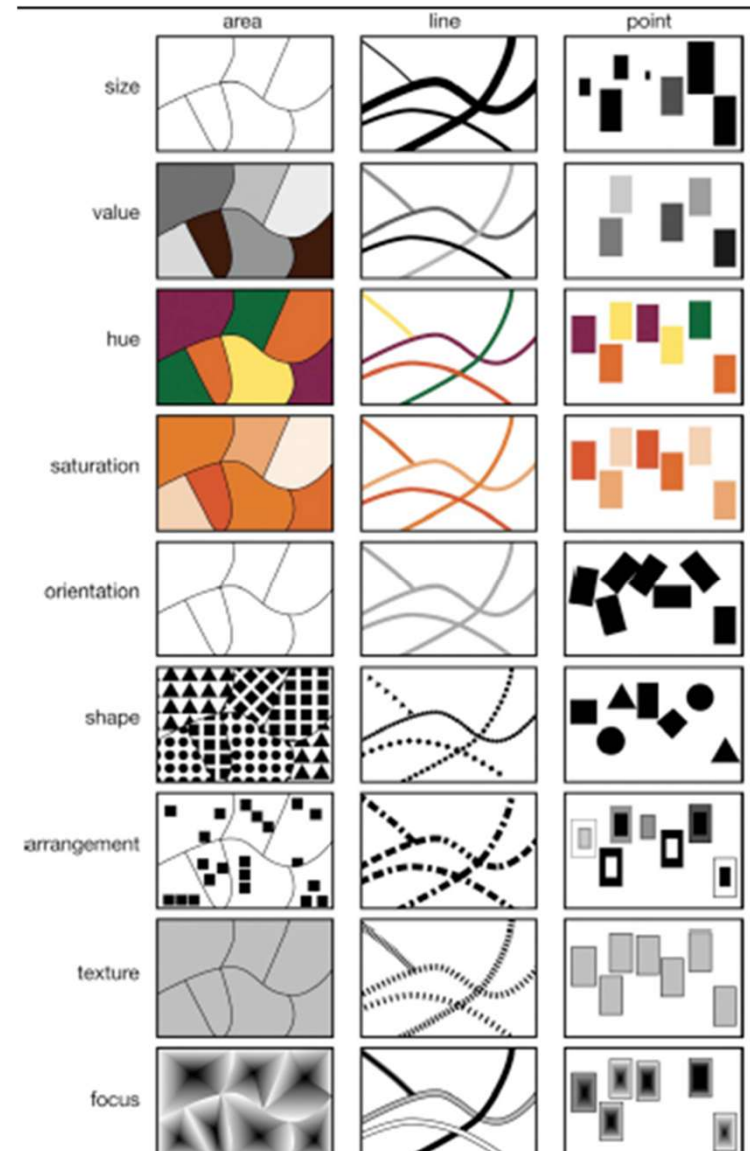


Portrait

Symbolization & visual variables

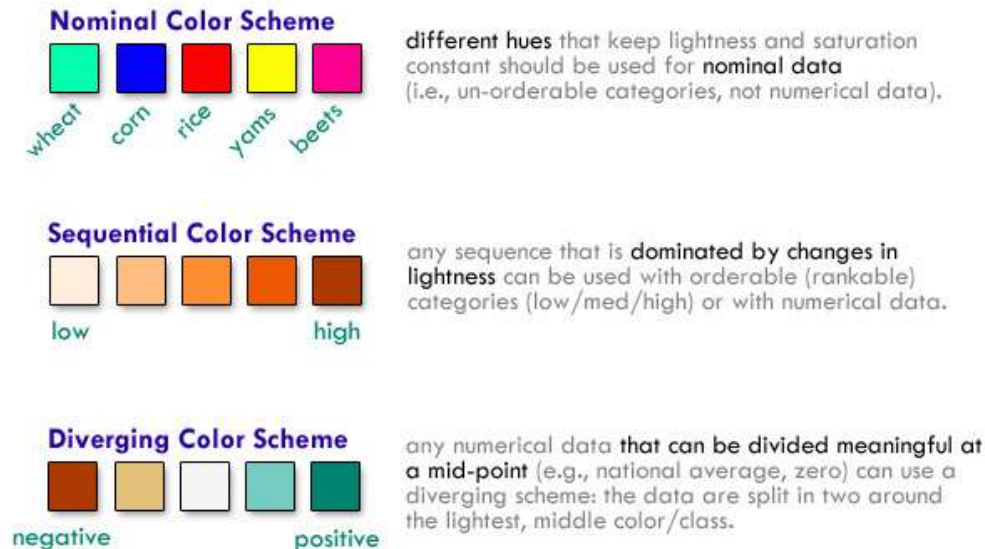
**Bertin's graphic primitives,
extended from seven to ten
variables (the variable
location is not depicted)**

Source: MacEachren 1994 (from Visualization in
Geographical Information Systems, Hearnshaw H.M. and
Unwin D.J. (eds.). Reproduced by permission of John Wiley &
Sons, Ltd.)



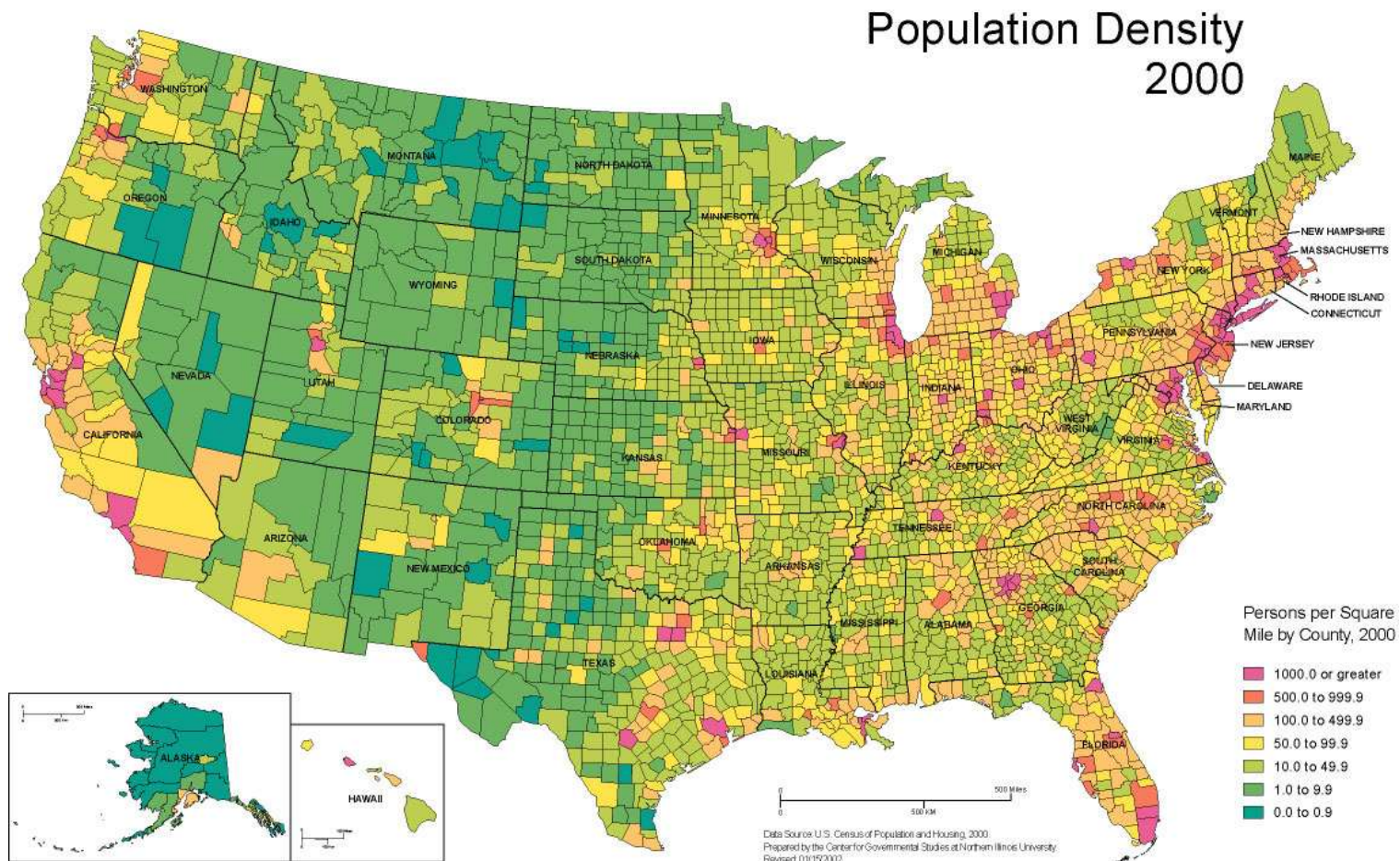
Color rules

- Type of thematic data determines your color scheme

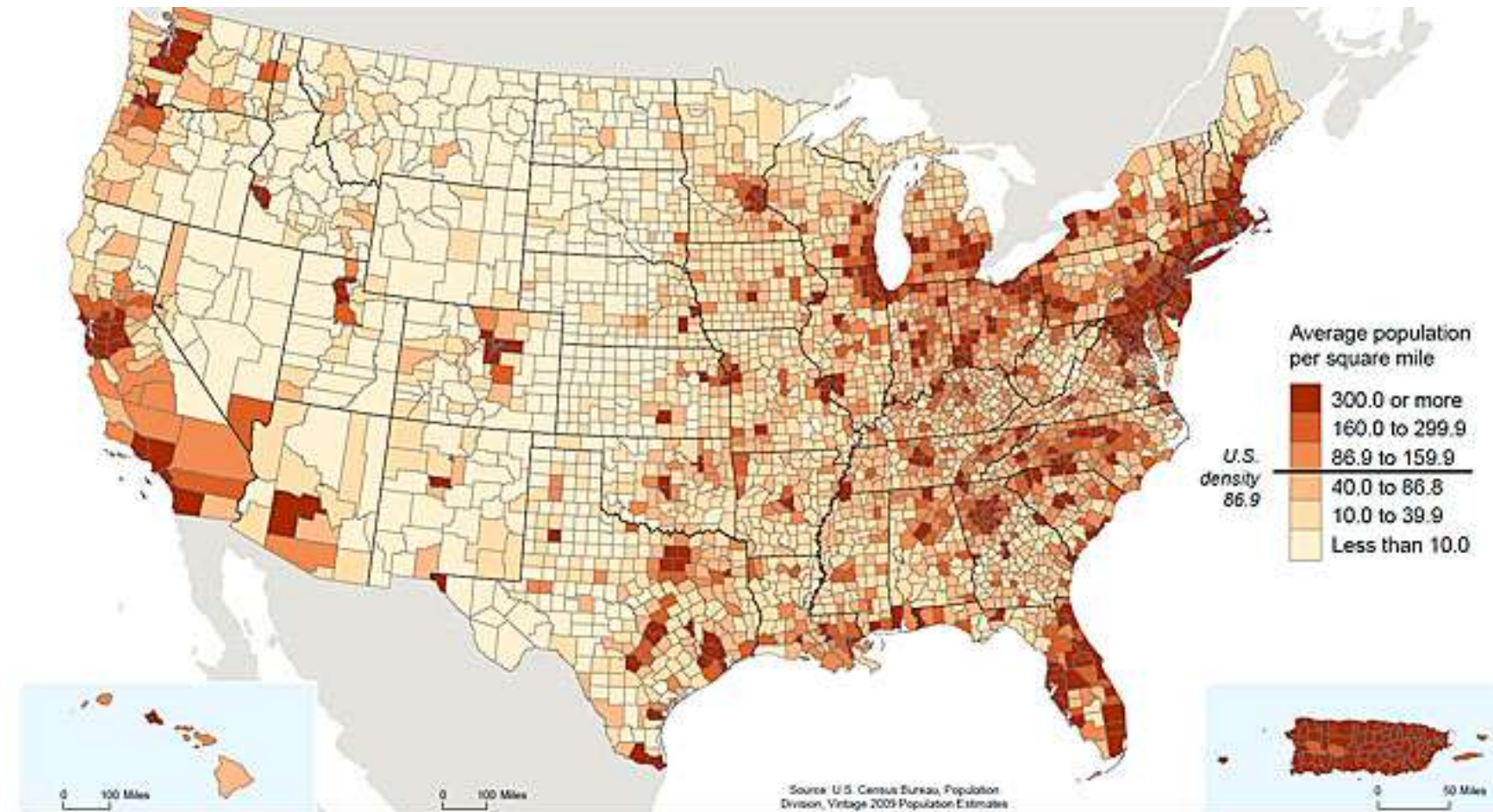


- Shading generally progresses from dark to light
 - Dark is associated with high values

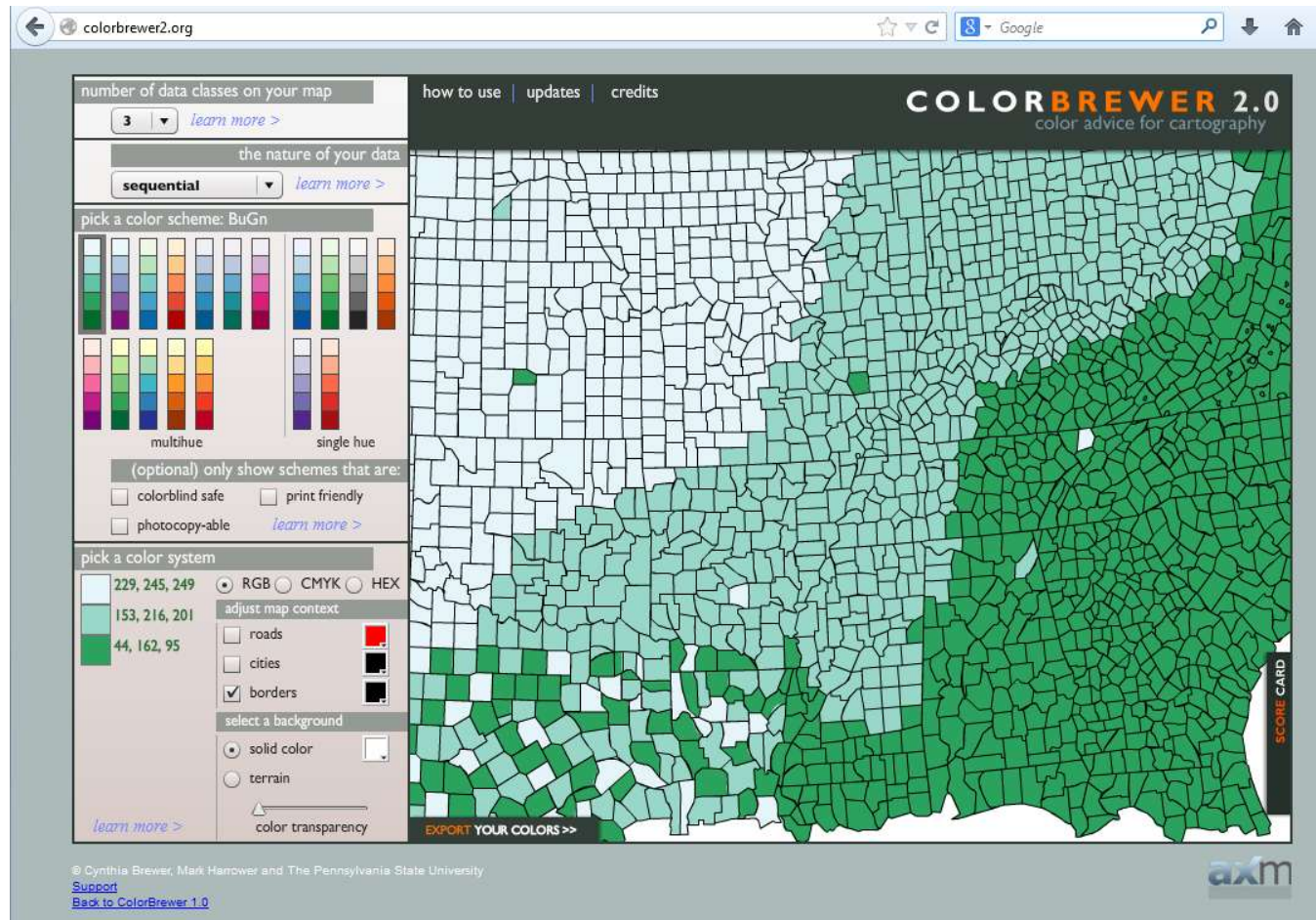
How do we process color?



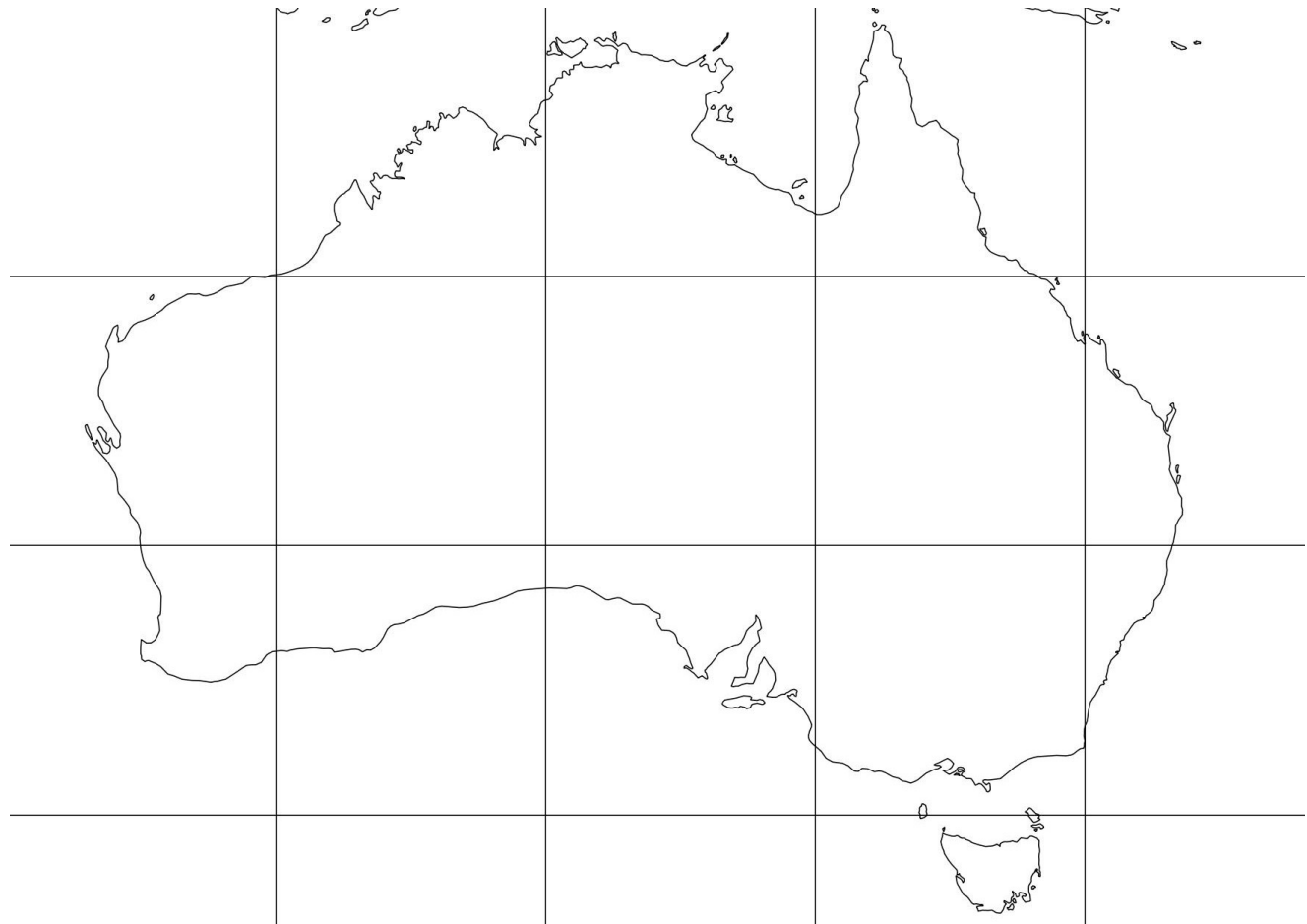
A population map of US counties



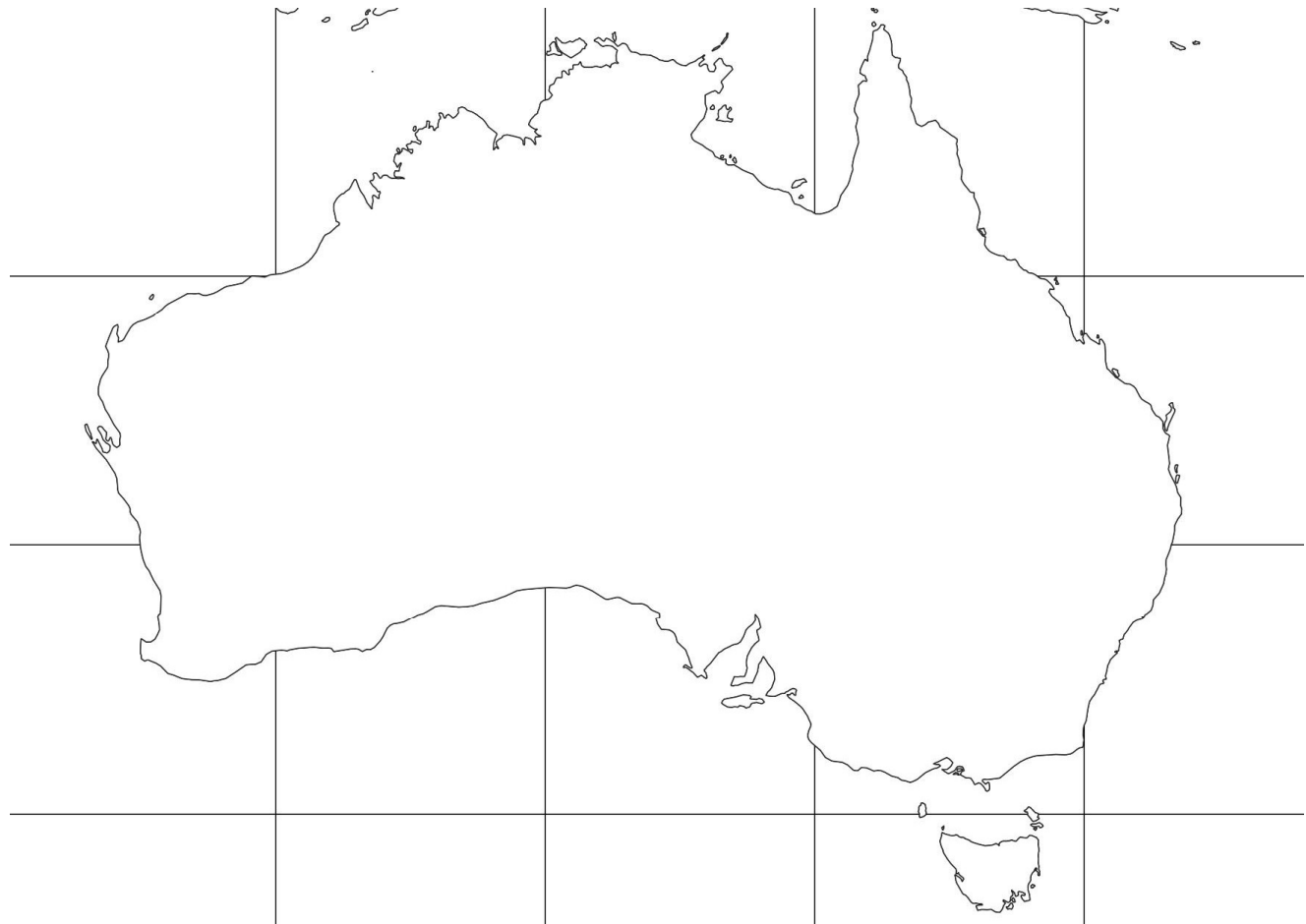
Color selection: one helpful tool



Line importance



Line importance



Line importance



Three main ways to “lie”

1. Selective truth
2. Cut-points and frivolous fills
3. Maps and bivariate *correlation*

